PPI Network Guided Driver Target Discovery

Computational Biology
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Content

1. Introduction
2. Method
3. Experimental results & Case study

Introduction

Problem Formulation

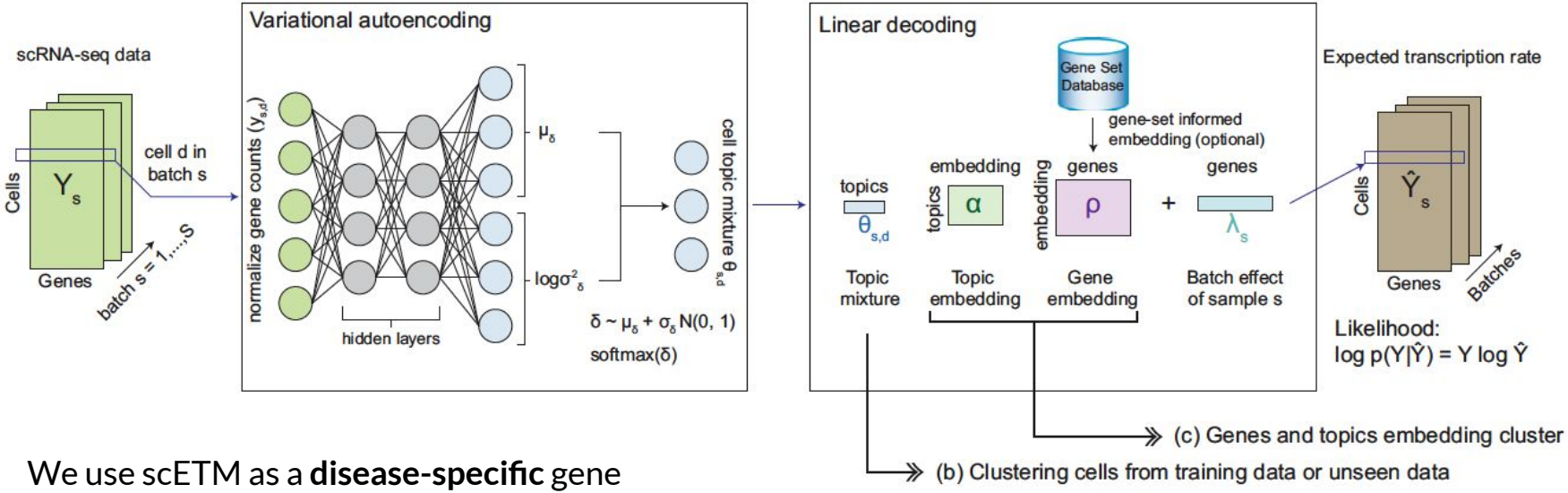
The **drug target discovery** task aims to mine possible gene targets for a given disease.

Protein-Protein Interation (PPI) : A graph containing interactions between proteins (genes), including shared function, binding, activation, inhibition, etc.

Input: Single-cell RNA-seq data, a list of known targets, **PPI network**

Output: a list of new targets

Related Work: scETM



We use scETM as a **disease-specific** gene feature extractor in our Framework

Zhao, Yifan, et al. "Learning interpretable cellular and gene signature embeddings from single-cell transcriptomic data." Nature Communications 12.1 (2021): 1-15.

Related Work: PPI-Guided Disease Protein Discovery

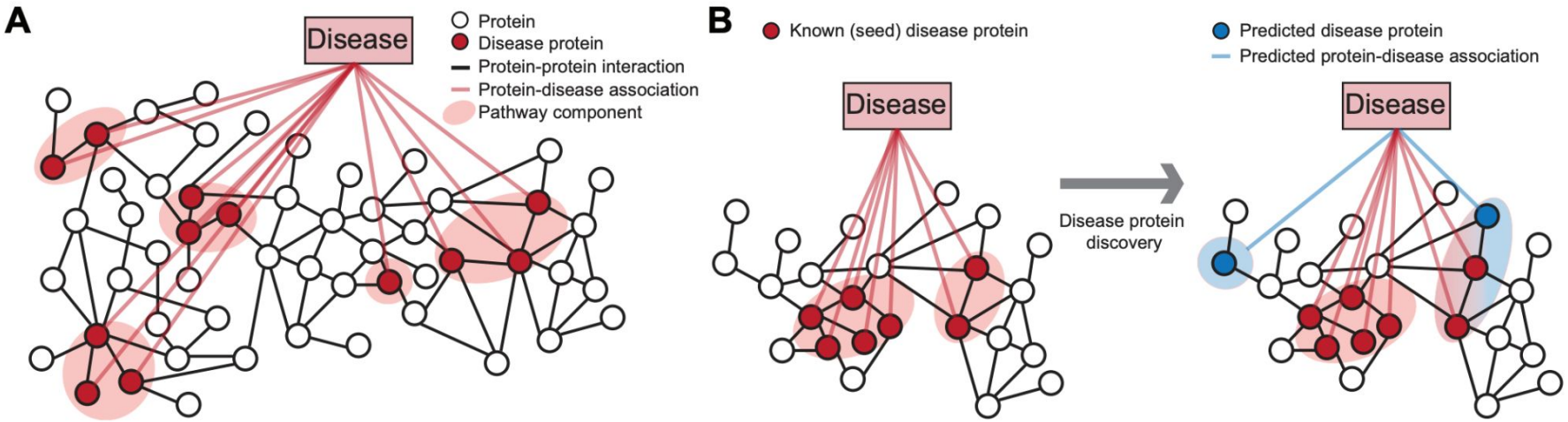


Image from: Agrawal, Monica, Marinka Zitnik, and Jure Leskovec. "Large-scale analysis of disease pathways in the human interactome." PACIFIC SYMPOSIUM ON BIOCOMPUTING 2018: Proceedings of the Pacific Symposium. 2018.

Method

Motivation

Problem:

Original pipeline does not consider interactions between genes when training cell and gene embeddings

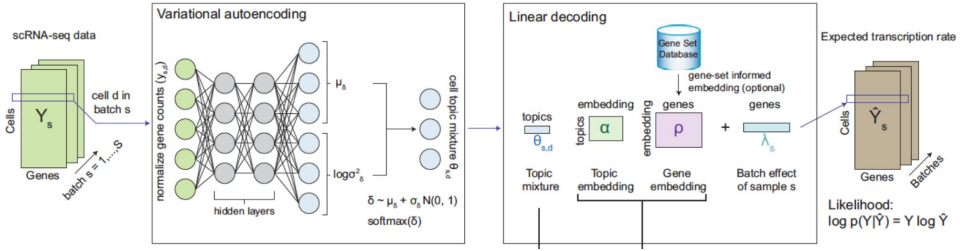
Motivation:

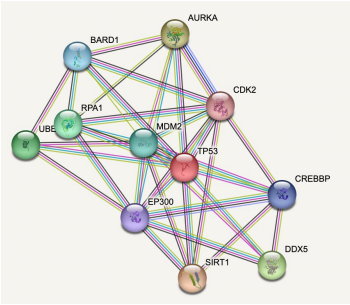
Simiar genes correspond to simiar targets

-> Add biological gene constraints to refine embeddings

Overall Schema

Gene Feature Extractor: sc-ETM

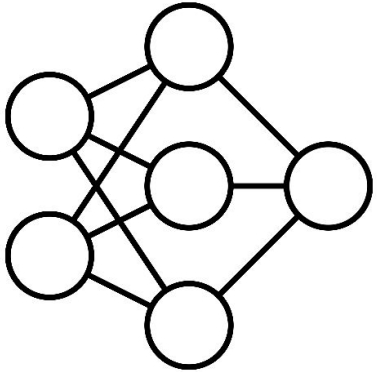




Extracted PPI Network

Predicted Gene Targets

Select Top K



Target Score Regression Head

0.5
0.7
...
0.8
0.2

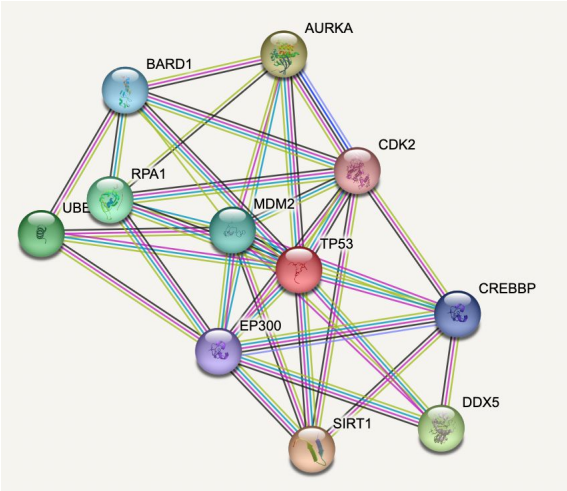
Settings for target discovery

Two steps for target discovery:

- Train a target score regressor on training set
- Pick the targets with highest predicted scores on test set

Our target discovery framework is designed for well-studied diseases, e.g. glioma in our case, where known target info is relatively sufficient.

PPI Network



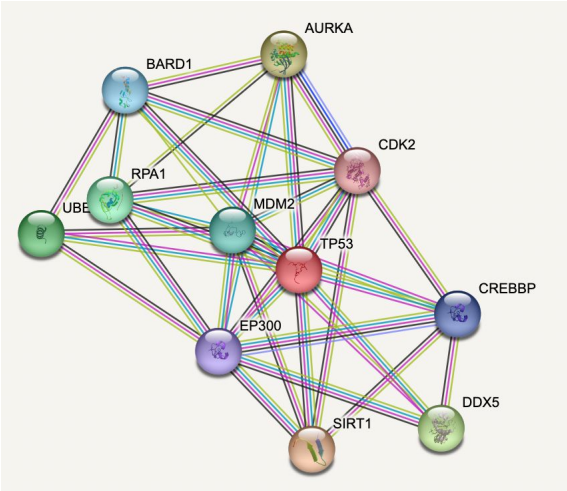
Filter with RNA-seq →

ARF5	PAK2	197
ARF5	RAB36	222
ARF5	RAPGEF1	181
ARF5	SUMO3	270
ARF5	ACAP2	767
ARF5	STX1B	155
ARF5	PXN	337
ARF5	SPRED3	158
ARF5	OTUB1	320
ARF5	ZDHHC3	247

STRING

- 6184316 links with confidence scores
- 12015 genes
- The edges are relations (shared function/binding/activation/inhibition…). We ignore specific meaning of each edge.

Regularization on PPI



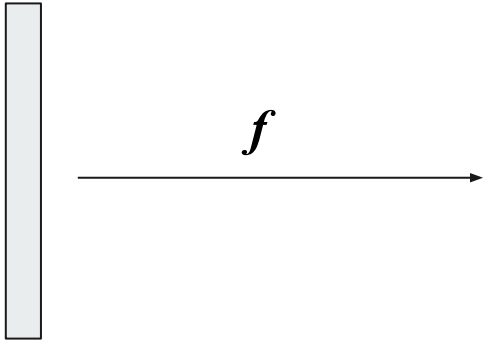
PPI graph $G = (V, E)$

Intuition:
Genes with strong biological correlation (according to the PPI network) should have similar embedding.

$$\mathcal{L}_{reg} = \sum_{e=(v_1,v_2)\in E} \underbrace{w_e}_{\text{Graph weights}} \cdot \left\| \underbrace{\rho_{v_1} - \rho_{v_2}}_{\text{Gene Embedding}} \right\|_2^2$$

$$\mathcal{L} = \mathcal{L}_{ori} + \lambda_{reg}\mathcal{L}_{reg}$$

Drug target discovery



Gene embedding ρ_v

Target score y

$$Min. \quad ||f(\rho_v) - y||$$

- We train a simple 2-layer MLP to model the transformation.
- Best model is selected with validation set and early stopping.

Results & Case Study

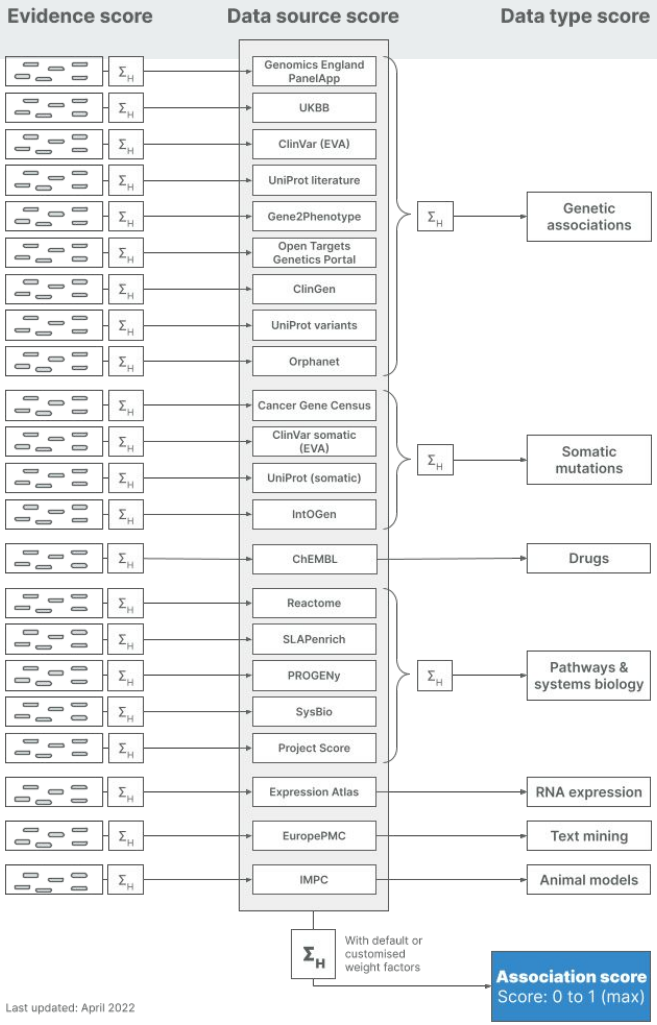
Dataset

- sc-RNA seq: Glioma transcriptional Cells (**GBM**, IDH-MUT)
  - Glioma is a type of tumor that occurs in brain and spinal cord
  - Single-cell data: 948 cells * 12916 genes
- Target Dataset: Open Targets Dataset
  - 7751 Glioma targets with target scores
  - 4699 overlapping genes with GBM
- Train / Valid / Test : 8/1/1

Chaligne, Ronan, et al. "Epigenetic encoding, heritability and plasticity of glioma transcriptional cell states." Nature genetics 53.10 (2021): 1469-1479.

Dataset

https://platform.opentargets.org/disease/EFO_0005543/associations

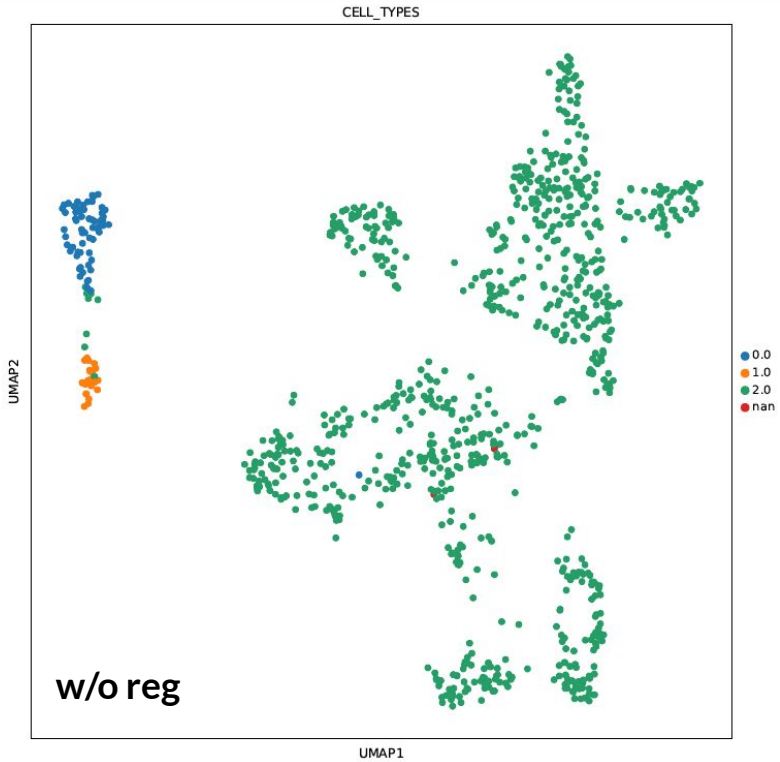


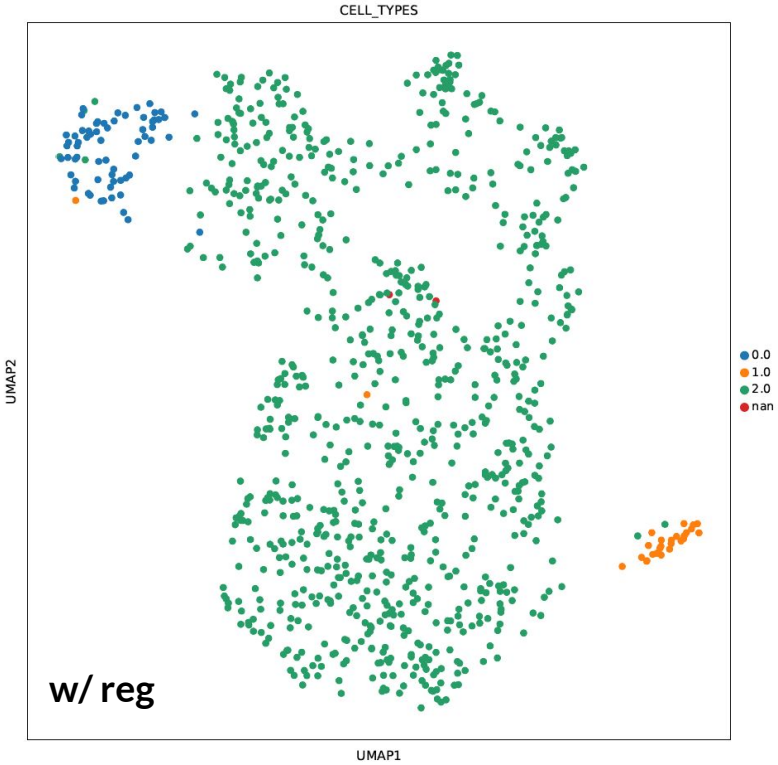
Results of clustering

- We use Leiden to cluster cells by their cell embeddings.
- During the clustering, we tried multiple resolution values and reported the result with the highest ARI for each method.

	ARI	NMI
w/o PPI guidance	0.9122	0.7735
w/ PPI guidance	**0.9260**	**0.8356**

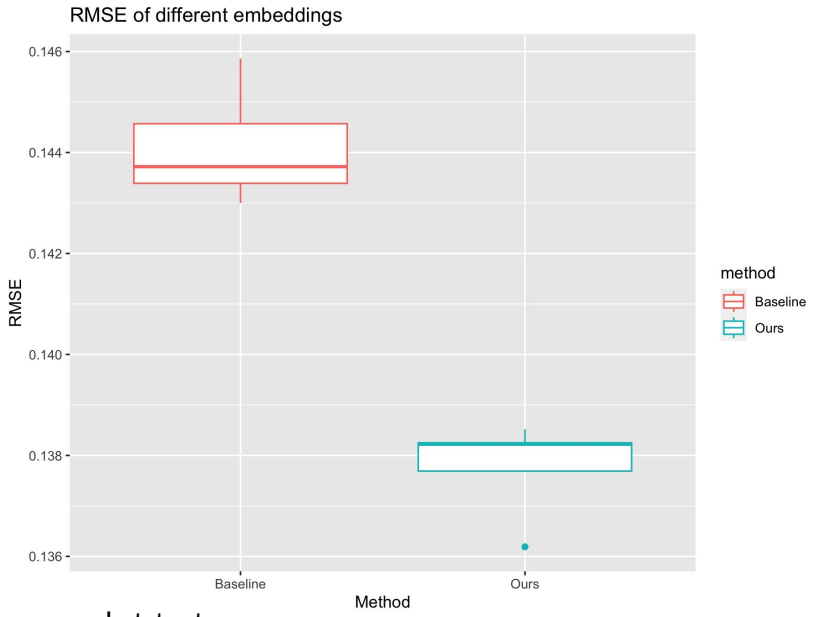
Visualization of clustering



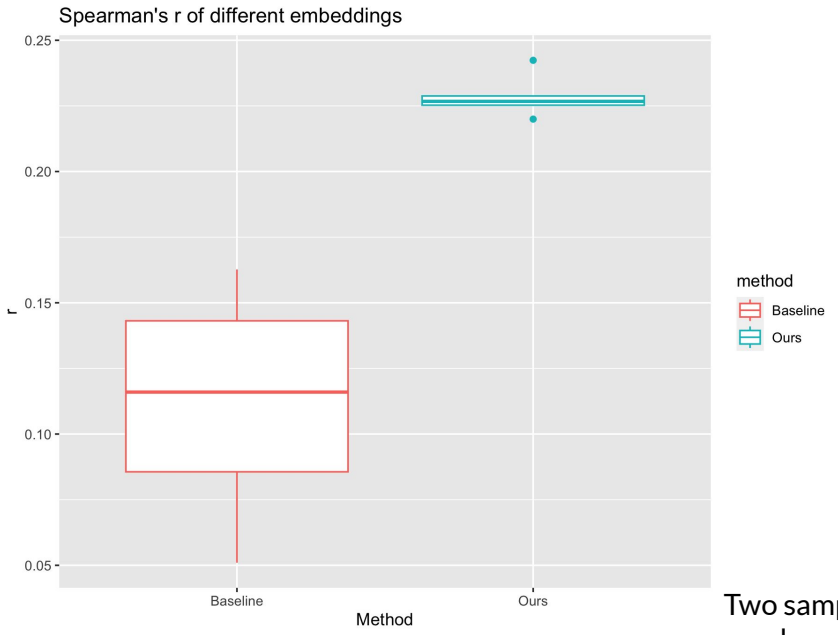


Results of target discovery

5 different seeds for each method



Two sample t-test:
p-value=1.458e-05



Two sample t-test:
p-value=.0037

Conclusion:

- Our method has lower RMSE and higher Spearman's r
- Our method has much lower variance

Results of target discovery

Top 10 genes on test set:

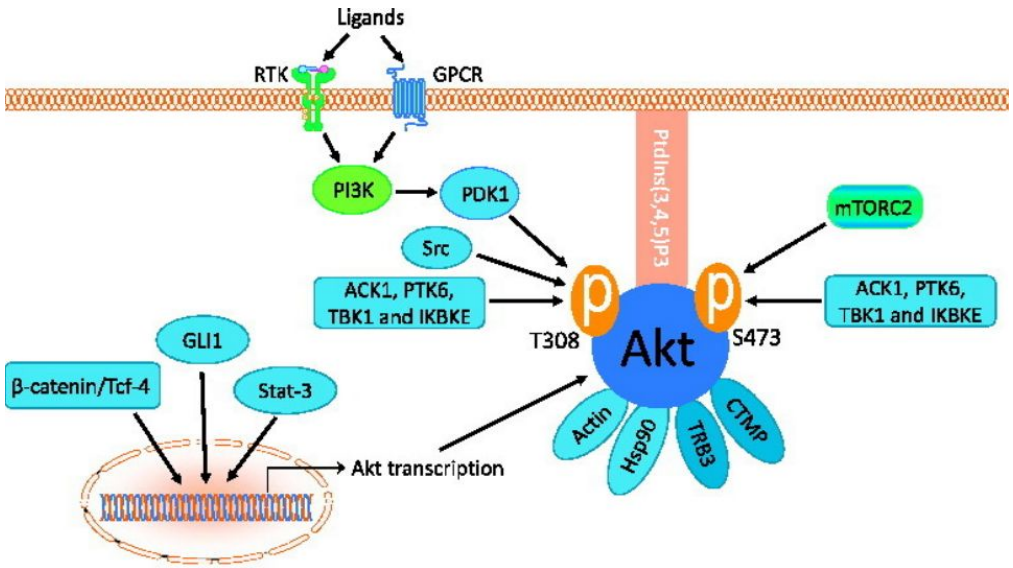
Gene	ZEB2	TCF4	MACF1	IGF2R	BIRC2	ALMS1	PYM1	B4GALT3	CIB1	C9orf72
Predicted score	.1778	.1801	.1837	.1931	.2250	.3816	.3137	.2981	.4250	.2922

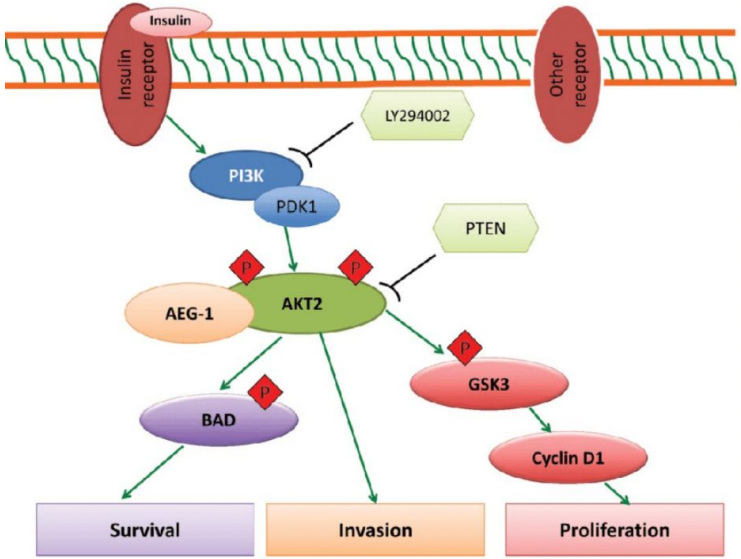
Gene	Expression	Significant pathway	Function verification
ZEB2	upregulate	Promoter of Epithelial-mesenchymal transition (EMT)	Knockdown by siRNA: proliferation↓, migration↓, invasion↓, apoptosis↑
TCF4	upregulate	driver gene in glioma, Wnt/β-catenin/Tcf pathway	Knockdown by siRNA: proliferation↓, , apoptosis↑
MACF1	predominantly express in glioblastoma	cross-linking actin to other cytoskeletal proteins bind to microtubules	Downregulation with RNAi:proliferation↓, migration↓,
IGF2R	overexpress	Cargo recognition for clathrin-mediated endocytosis, Clathrin-mediated endocytosis	
BIRC2	overexpress	Ub-specific processing proteases, Deubiquitination, Apoptosis, Programmed Cell Death	

Gene	Expression	Significant pathway	Function verification
ALMS1	upregulate	G2/M Transition, Mitotic G2-G2/M phases	
PYM1		Key regulator of the exon junction complex	
B4GALT3	upregulate	N-linked oligosaccharides β-catenin/vimentin pathway	Knockdown by siRNA: proliferation↓, invasion↓
CIB1	overexpress	oncogenic signalling by Ras and SK1	
C9orf72		Component of the C9orf72-SMCR8 complex	

Case Study

Tcf-4 play an important role in glioma mediated by AKT2

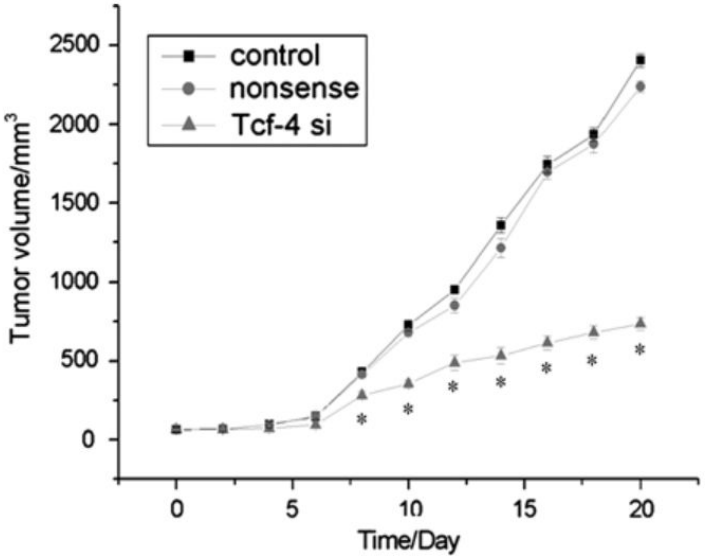




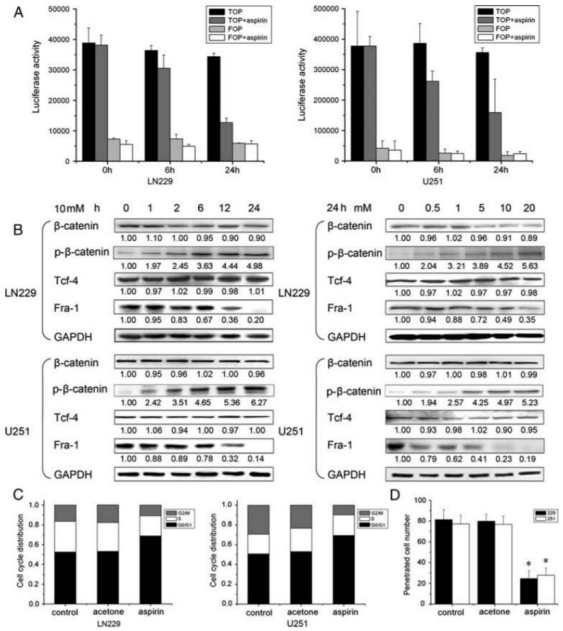
Prasanna Abeyrathna and Yunchao Su. The critical role of akt in cardiovascular function. Vascular pharmacology, 74:38–48, 2015.

Case Study

Reduction of Tcf-4 inhibits glioma growth in vivo



Effect of β-catenin/Tcf-4 activity on glioma cell proliferation and invasion



Junxia Zhang, Kai Huang, Zhendong Shi, Jian Zou, Yingyi Wang, Zhifan Jia, Anling Zhang, Lei Han, XiaoYue, Ning Liu, et al. High β-catenin/tcf-4 activity confers glioma progression via direct regulation of akt2 gene expression. Neuro-oncology, 13(6):600–609, 2011.7

Conclusion

- We design a PPI network guided driver target discovery framework.
- Experiments are conducted on the sc-RNA glioma tumor data, which show the effectiveness of incorporating PPI network guidance in target score regression.
- Case studies on our discovered target list further verify the capability of our designed framework.